The Algorithmic Monoculture Paradox: Balancing Individual Benefit and Collective Error in Foundation-Model-Mediated Networks

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Abstract

The "Wisdom of Crowds" principle establishes that distributed networks can achieve high collective accuracy through idiosyncratic error cancellation. Today, the rapid integration of foundation models into decision workflows fundamentally alters the epistemic structure of networked deliberation. While incorporating foundation-model advice produces substantial reductions in individual expected squared error—reducing individual error by over 12-fold in task-persistent settings (from 225.0 to 18.37) and up to 46-fold under dynamic sampling (to 4.89)—correlated machine errors ($\rho = 0.90$) accumulate in the collective consensus. Combining the Krogh-Vedelsby / Page decomposition with iterated DeGroot dynamics validated across 193,000 unique Monte Carlo runs ($N = 1,000$ agents, $T = 30$ rounds; 199,000 total executed trials), we analyze this tension. In idealized orthogonal crowds ($\rho_h = 0$), monoculture causes collective error to escalate 57-fold (from 0.327 ± 0.009 to 18.648 ± 0.468), though crowd error remains 12× lower than unassisted individual error. Crucially, because initial reliance reduces error ($\frac{\partial E}{\partial \alpha}\big|_{\alpha=0} = -2T \cdot \mathrm{MSE}_{\mathrm{human}} < 0$), an interior optimum exists even for orthogonal humans; when humans exhibit correlated estimation errors ($\rho_h > 0$), this negative gradient is amplified, establishing a pronounced U-shaped trade-off. Socially optimal per-step reliance ($\alpha^* \in [0.008, 0.056]$, cumulative $w_{\mathrm{cum}} \in [20\%, 82\%]$) reduces collective error by 20%–83%. When decentralized agents adopt isolated reliance benchmarks under correlation neglect, the stylized Epistemic Reliance Overshoot Ratio spans $\mathcal{R}_{\mathrm{overshoot}} \in [1.2, 4.1]$, reaching up to 8.0× at complete per-step reliance ($\alpha_{\mathrm{step}} = 1.0$). When human correlation exceeds $\rho_h^* \approx 8.1\%$ (exact 8.085%), pure AI ($\alpha_{\mathrm{step}} = 1.0$) achieves lower error than unassisted humans, rendering excessive reliance an epistemic overshoot rather than a degradation. We prove analytically that uniform external advice passes invariant through row-stochastic networks ($\mathbf{W1} = \mathbf{1}$) under homogeneous reliance, and show that multi-model competition ($\rho = 0.50$) provides limited protection (error ≈ 11.6 vs. 18.6 under monopoly monoculture).

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1 Introduction

For over a century, mathematical statistics and social epistemology have recognized that collective intelligence emerges from cognitive diversity (Galton, 1907; Krogh and Vedelsby, 1995; Page, 2007; Lorenz et al., 2011; Becker, Brackbill, and Centola, 2017). The aggregation of sufficiently diverse and approximately unbiased individual judgments can produce a collective estimate that is more accurate than the average individual judgment.

The rapid worldwide deployment of generative foundation models (Large Language Models, or LLMs) has disrupted this epistemic architecture (Bommasani et al., 2022). Today, broad populations of professionals, researchers, students, and autonomous software agents consult a small cluster of centralized frontier models for decision support, text synthesis, and heuristic guidance.

Under the parameterized error distributions considered here, incorporating foundation-model advice produces substantial reductions in individual expected squared error: under the specified error distribution, AI advice reduces expected squared error and provides strong local variance reduction. However, evaluating AI assistance exclusively through individual benchmarks commits a fallacy of composition: assuming that what improves individual accuracy necessarily benefits the collective. Building on the theory of algorithmic monoculture formalized by Kleinberg and Raghavan (2021), when independent decision-makers adopt identical or highly correlated algorithmic rules, the system exhibits Algorithmic Monoculture.

Recent empirical audits confirm that commercial frontier foundation models exhibit substantial shared failure modes across diverse architectures: for instance, Kim et al. (2025) evaluate over 350 LLMs on a popular leaderboard benchmark and observe that distinct models agree on identical incorrect multiple-choice answers approximately 60% of the time conditional on both failing. While discrete classification overlap differs from continuous scalar error correlation, it highlights that algorith-

mic outputs fail to act as independent signals. Meanwhile, Wu, Black, and Chandrasekaran (2025) suggest that post-training alignment processes may contribute to generative monoculture by homogenizing model output distributions. Concurrently, behavioral studies demonstrate that generative assistance can reduce the collective diversity of novel content across human creators (Doshi and Hauser, 2024). In real-world environments, humans do not interact with algorithms in isolation; they deliberate, exchange beliefs, and form consensus across structured communication networks (Watts and Strogatz, 1998; Barabási and Albert, 1999; Golub and Jackson, 2010; Zollman, 2007). In human crowds, social influence, shared heuristics, and common media sources induce baseline error correlation $\rho_h > 0$ (Lorenz et al., 2011; Soll and Larrick, 2009; Becker, Brackbill, and Centola, 2017); in our analysis, we evaluate illustrative stylized regimes of modest to moderate correlated human error ($\rho_h \in [0.02, 0.10]$).

This paper addresses three central open questions:

1. How does algorithmic reliance (α) interact with human cognitive correlation (ρ_h) and cross-agent AI correlation (ρ) in networked crowds?
2. Under what conditions does foundation model advice improve versus degrade collective accuracy?
3. Why does an isolated decision-maker's private reliance benchmark overshoot the socially optimal reliance level in deliberating networks?

Through a unified analytical framework verified by 193,000 unique high-throughput Monte Carlo simulation runs across $1,000$-agent networks ($K = 1,000$ independent trials per condition; 199,000 total executed trials), we combine the Krogh-Vedelsby / Page ensemble error decomposition (Krogh and Vedelsby, 1995; Page, 2007) with iterated DeGroot consensus dynamics under persistent external influence (DeGroot, 1974; Friedkin and Johnsen, 1990; Acemoglu et al., 2013). Our findings demonstrate that algorithmic monoculture creates an epistemic dilemma: while moderate AI reliance anchors human crowds subject to correlated estimation errors, private optimization benchmarks encourage reliance levels that drive the network into a regime of collective epistemic overshoot.

2 Theoretical Framework

2.1 Consensus Dynamics and Ambiguity Decomposition

Consider an objective state of the world $\theta \in \mathbb{R}$ estimated by a network of N deliberating agents. Let $\mathbf{x}(t) = [x_1(t), \ldots, x_N(t)]^\top \in \mathbb{R}^N$ denote the vector of agent beliefs at timestep t. Agent beliefs evolve through iterative local communication combined with repeated consultation of external foundation model advice:

$$\mathbf{x}(t+1) = (1-\alpha)\mathbf{W}\mathbf{x}(t) + \alpha\hat{\boldsymbol{\theta}}_M(t) \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{N\times N}$ is an aperiodic, irreducible row-stochastic trust matrix ($\mathbf{W}\mathbf{1} = \mathbf{1}$), $\alpha \in [0,1]$ is the per-step algorithmic reliance weight, and $\hat{\boldsymbol{\theta}}_M(t)$ is the external advice vector provided to the agents at round t.

To formalize the interplay between individual error and collective judgment, we employ the Krogh-Vedelsby / Page diversity decomposition (Krogh and Vedelsby, 1995; Page, 2007). Let $c(t) = \frac{1}{N}\sum_{i=1}^N x_i(t) = \frac{1}{N}\mathbf{1}^\top\mathbf{x}(t)$ denote the unweighted arithmetic crowd mean. At any realization, the collective squared error decomposes pointwise as:

$$(c(t)-\theta)^2 = \frac{1}{N}\sum_{i=1}^N (x_i(t)-\theta)^2 - \text{Diversity}(t) \tag{2}$$

where $\frac{1}{N}\sum_{i=1}^N (x_i(t)-\theta)^2$ is individual squared error and $\text{Diversity}(t) = \frac{1}{N}\sum_{i=1}^N (x_i(t)-c(t))^2$ is empirical belief variance; taking expectations across trials yields the MSE decomposition $\mathbb{E}[(c(t)-\theta)^2] = \frac{1}{N}\sum_{i=1}^N \mathbb{E}[(x_i(t)-\theta)^2] - \mathbb{E}[\text{Diversity}(t)]$.

Following parameterized correlated error formulations (Kleinberg and Raghavan, 2021), external advice to agent i at round t is modeled as:

$$\hat{\theta}_{M,i}(t) = \theta + \mu_{AI} + \sqrt{\rho}\,\eta_{\text{shared}}(t) + \sqrt{1-\rho}\,\eta_{\text{indiv},i}(t) \tag{3}$$

where μ_{AI} is the systematic directional AI mean error, $\eta_{\text{shared}}(t) \sim \mathcal{N}(0, \sigma_{AI}^2)$ is a common error shock shared across agents, and $\eta_{\text{indiv},i}(t) \sim \mathcal{N}(0, \sigma_{AI}^2)$ is an agent-specific residual error capturing variations in individual prompt formulations, context windows, or stochastic temperature sampling. The parameter $\rho \in [0,1]$ represents the cross-agent AI error correlation. We explicitly specify the underlying statistical independence structure: (i) $\eta_{h,\text{shared}} \perp \eta_{h,i}$ and $\eta_{h,i} \perp \eta_{h,j}$ for $i \neq j$; (ii) in dynamic noise, $\eta_{\text{shared}}(t) \perp \boldsymbol{\eta}_{\text{indiv}}(t)$, with residuals independent across agents and rounds, and shared shocks independent across rounds; (iii) in static noise, shared and residual shocks are mutually independent but fixed over rounds; and (iv) human priors and AI errors are mutually independent in the baseline model ($\boldsymbol{\eta}_h \perp \boldsymbol{\eta}_{AI}(t)$).

2.2 Finite-Time Consensus Dynamics and Topology Invariance

In networked deliberation, collective judgment is measured by the arithmetic crowd mean $c(t) = \frac{1}{N}\sum_{i=1}^N x_i(t) = \frac{1}{N}\mathbf{1}^\top\mathbf{x}(t)$. To characterize $c(t)$ in finite time across arbitrary network topologies, we define the finite-time projection vector:

$$\mathbf{q}_k^\top \equiv \frac{1}{N}\mathbf{1}^\top\mathbf{W}^k \quad (k \geq 0) \tag{4}$$

with initial condition $\mathbf{q}_0^\top = \frac{1}{N}\mathbf{1}^\top$. Because $\mathbf{W}$ is row-stochastic ($\mathbf{W1} = \mathbf{1}$), every matrix power satisfies $\mathbf{W}^k\mathbf{1} = \mathbf{1}$. Consequently, the projection vector preserves total probability mass identically at every round:

$$\mathbf{q}_k^\top \mathbf{1} = \frac{1}{N}\mathbf{1}^\top \mathbf{W}^k \mathbf{1} = \frac{1}{N}\mathbf{1}^\top \mathbf{1} = 1 \quad \forall k \geq 0 \tag{5}$$

Unrolling the belief evolution Equation 1 from $t = 0$ to horizon T yields:

$$\mathbf{x}(T) = (1-\alpha)^T \mathbf{W}^T \mathbf{x}(0) + \sum_{k=0}^{T-1} \alpha(1-\alpha)^k \mathbf{W}^k \hat{\boldsymbol{\theta}}_M (T-1-k) \tag{6}$$

Left-multiplying by $\frac{1}{N}\mathbf{1}^\top$ produces the exact finite-time arithmetic crowd mean:

$$c(T) = (1-\alpha)^T \mathbf{q}_T^\top \mathbf{x}(0) + \sum_{k=0}^{T-1} \alpha(1-\alpha)^k \mathbf{q}_k^\top \hat{\boldsymbol{\theta}}_M (T-1-k) \tag{7}$$

Decomposing external advice into uniform shared advice and individual residual noise:

$$\hat{\boldsymbol{\theta}}_M(t) = \theta_0(t)\mathbf{1} + \sqrt{1-\rho}\,\boldsymbol{\eta}_{\text{indiv}}(t) \tag{8}$$

where $\theta_0(t) = \theta + \mu_{AI} + \sqrt{\rho}\,\eta_{\text{shared}}(t)$. By the stochastic identity (Equation 5), $\mathbf{q}_k^\top(\theta_0(t)\mathbf{1}) = \theta_0(t)(\mathbf{q}_k^\top \mathbf{1}) = \theta_0(t)$ for every round k. Consequently, the uniform shared advice mode decouples identically from the network topology $\mathbf{W}$ at every finite horizon T. The cumulative weight on external advice over T deliberation rounds defines cumulative reliance:

$$\boxed{w_{\text{cum}} = 1 - (1-\alpha)^T \equiv b_T} \tag{9}$$

Because $\sum_{k=0}^{T-1} \alpha(1-\alpha)^k = b_T$, the expected mean bias is strictly topology-invariant: $\mathbb{E}[c(T) - \theta] = \mu_{AI} b_T$.

Meanwhile, let $\mathbf{P} = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ denote the centering projector such that $\text{Diversity}(t) = \frac{1}{N}\|\mathbf{P}\mathbf{x}(t)\|_2^2$. Because $\mathbf{W}^k\mathbf{1} = \mathbf{1}$ and $\mathbf{P1} = \mathbf{0}$, the uniform shared advice mode vanishes ($\mathbf{PW}^k\mathbf{1} = \mathbf{0}$). Under dynamic sampling, the exact finite-time belief diversity at horizon T evaluates to:

$$\begin{aligned}\mathbb{E}[\text{Diversity}(T)] = {} & \frac{(1-\alpha)^{2T}(1-\rho_h)\sigma^2_{\text{human}}}{N} \\ & \times \text{Tr}\big(\mathbf{P}\mathbf{W}^T(\mathbf{W}^T)^\top \mathbf{P}\big) \\ & + \frac{\alpha^2(1-\rho)\sigma^2_{AI}}{N} \sum_{k=0}^{T-1} (1-\alpha)^{2k} \\ & \times \text{Tr}\big(\mathbf{P}\mathbf{W}^k(\mathbf{W}^k)^\top \mathbf{P}\big)\end{aligned} \tag{10}$$

where the first term represents residual human disagreement (evaluating to 1.474 at $\alpha = 0$ on Watts-Strogatz due to incomplete spectral mixing at $T = 30$, matching simulated 1.454 ± 0.011) and the second captures dispersion from idiosyncratic AI noise. As $T \to \infty$, $(1-\alpha)^{2T} \to 0$, recovering the steady-state limit $\lim_{t\to\infty} \mathbb{E}[\text{Diversity}(t)] = \frac{\alpha^2(1-\rho)\sigma^2_{AI}}{N} \sum_{k=0}^{\infty}(1-\alpha)^{2k}\text{Tr}(\mathbf{P}\mathbf{W}^k(\mathbf{W}^k)^\top\mathbf{P})$, bounded strictly by $\mathbb{E}[\text{Diversity}(\infty)] \leq \frac{\alpha}{2-\alpha}(1-\rho)\sigma^2_{AI}$ for $0 < \alpha < 1$.

2.3 Temporal Persistence: Dynamic vs. Static AI Noise

In multi-round deliberation, the temporal accumulation of algorithmic error depends on whether model errors are stochastic across queries or fixed for a given task:

1. Dynamic AI Noise (Temporal Stochasticity): In this regime, agents refresh queries at each round via stochastic temperature draws or track an ongoing information stream where α acts as an exponential moving average (EMA) filter. Here, the shared error $\eta_{\text{shared}}(t)$ is i.i.d. across rounds, and idiosyncratic AI errors are independent across both agents and successive rounds. Taking the expectation of $(c(T)-\theta)^2$ from Equation 7 yields the exact finite-time collective squared error:

$$\begin{aligned}E_{\text{dynamic}} = {} & (1-\alpha)^{2T}\text{MSE}_{\text{human}}(\rho_h, T) \\ & + E_{\text{AI}}^{\text{dyn}}(\alpha, \rho, T)\end{aligned} \tag{11}$$

where the exact AI-induced error contribution evaluates to:

$$\begin{aligned}E_{\text{AI}}^{\text{dyn}} = {} & \mu_{AI}^2 b_T^2 + \frac{\alpha(1-(1-\alpha)^{2T})}{2-\alpha}\rho\sigma^2_{AI} \\ & + \alpha^2(1-\rho)\sigma^2_{AI}\sum_{k=0}^{T-1}(1-\alpha)^{2k}\|\mathbf{q}_k\|^2\end{aligned} \tag{12}$$

and $\text{MSE}_{\text{human}}(\rho_h, T) = \sigma_h^2[\rho_h + (1-\rho_h)\|\mathbf{q}_T\|^2]$ with $\sigma_h^2 = 225.0$. As $T \to \infty$, E_{dynamic} converges to $\mu_{AI}^2 + \frac{\alpha\rho\sigma^2_{AI}}{2-\alpha} + \frac{\alpha(1-\rho)\sigma^2_{AI}}{2-\alpha}\sum_i \pi_i^2$.

2. Static AI Noise (Task-Persistent Error): In this task-persistent regime, each agent's task-specific AI error realization remains fixed across deliberation rounds ($\hat{\boldsymbol{\theta}}_M(t) \equiv \hat{\boldsymbol{\theta}}_M$ with $\eta_{\text{shared}}(t) \equiv \eta_{\text{shared}}$ and $\boldsymbol{\eta}_{\text{indiv}}(t) \equiv \boldsymbol{\eta}_{\text{indiv}}$ for all t), where the shared component captures common systematic estimation errors across agents and the residual component captures agent-specific deviations. Unrolling Equation 7 with time-invariant advice yields:

$$\begin{aligned}E_{\text{static}} = {} & (1-\alpha)^{2T}\text{MSE}_{\text{human}}(\rho_h, T) \\ & + E_{\text{AI}}^{\text{stat}}(\alpha, \rho, T)\end{aligned} \tag{13}$$

where:

$$E_{\text{AI}}^{\text{stat}}(\alpha, \rho, T) = b_T^2\left(\mu_{AI}^2 + \rho\sigma^2_{AI}\right) + (1-\rho)\sigma^2_{AI}\|\mathbf{v}_T\|_2^2 \tag{14}$$

with $\mathbf{v}_T^\top \equiv \sum_{k=0}^{T-1} \alpha(1-\alpha)^k \mathbf{q}_k^\top$. In the static regime, multi-round querying provides no temporal variance dampening, driving collective error directly toward the monoculture ceiling ($\mu_{AI}^2 + \rho\sigma_{AI}^2 \approx 18.40$ at $\rho = 0.90$) for any $\alpha > 0$ as T grows large.

Topology Invariance and Stationary Surrogation. The common-mode AI error is exactly topology-invariant, while the complete finite-time collective error exhibits only small topology-dependent residual differences under the parameter settings considered. As $k \to \infty$, the finite-time projection vector converges asymptotically to the stationary distribution, implying that $\|\mathbf{q}_k\|_2^2 \to \|\boldsymbol{\pi}\|_2^2 = \sum_i \pi_i^2$. We therefore use the stationary quantity as a numerical approximation when its difference from the finite-time projection is negligible. In our finite-horizon setting ($T = 30$), $\|\mathbf{q}_k\|_2^2$ converges rapidly toward this limit: on Watts-Strogatz ($\lambda_2 \approx 0.984$, spectral relaxation time $\tau_{\text{rel}} = \frac{1}{1-\lambda_2} \approx 62$ rounds), $\|\mathbf{q}_k\|_2^2$ increases smoothly from $\|\mathbf{q}_0\|_2^2 = 1/N = 0.001000$ to $\|\mathbf{q}_{30}\|_2^2 \approx 0.001015$ (within 10^{-6} of $\sum \pi_i^2 = 0.001016$); on Erdős-Rényi ($\lambda_2 \approx 0.64$, $\tau_{\text{rel}} \approx 2.8$) $\|\mathbf{q}_{30}\|_2^2 \approx 0.001018$; and on Barabási-Albert ($\lambda_2 \approx 0.86$, $\tau_{\text{rel}} \approx 7.1$) $\|\mathbf{q}_{30}\|_2^2 \approx 0.002474$ (matching the representative graph's $\sum \pi_i^2 \approx 0.00248$; across 100 independent BA graph ensembles, $\sum \pi_i^2 = 0.00238 \pm 0.00013$ standard deviation).

Formally, letting $\delta_k \equiv \left|\|\mathbf{q}_k\|_2^2 - \|\boldsymbol{\pi}\|_2^2\right|$, the discrepancy between exact finite-time error $E_c(T)$ and stationary surrogate $E_\pi(T)$ under dynamic noise is bounded by $|\Delta E| \leq \sigma_{\text{human}}^2(1-\rho_h)(1-\alpha)^{2T}\delta_T + \alpha^2(1-\rho)\sigma_{AI}^2 \sum_{k=0}^{T-1}(1-\alpha)^{2k}\delta_k$. On Watts-Strogatz and Erdős-Rényi graphs, $\max_k \delta_k \leq 1.94 \times 10^{-5}$ ($\delta_{30} \leq 3.1 \times 10^{-7}$), guaranteeing that $|\Delta E| < 0.00031$ across all $\rho \in [0,1]$ and all $\alpha \in [0.01, 1.0]$ ($|\Delta E| < 3.1 \times 10^{-5}$ at $\rho = 0.90$). On Barabási-Albert scale-free graphs, spectral contraction yields weighted sum $\sum_{k=0}^{T-1}(1-\alpha)^{2k}\delta_k \approx 3.01 \times 10^{-3}$ at $\alpha = 0.10$, bounding $|\Delta E| < 0.00005$ at $\alpha = 0.10, \rho = 0.90$ (< 0.00049 across all $\rho \geq 0$), and at complete reliance $\alpha = 1.0$, bounding $|\Delta E| \leq 0.0024$ under $\rho = 0.90$, ≤ 0.012 under $\rho = 0.50$, and ≤ 0.024 under $\rho = 0.00$ (static analyses evaluate exact finite-time $\mathbf{v}_T$ directly). Consequently, finite-time differences are negligible across all practical regimes (e.g., at $\alpha = 0.10, \rho = 0.90$, $E_c^{\text{dyn}} = 4.4251$ on WS/ER, 4.4258 on BA; stationary $E_\pi^{\text{dyn}} = 4.4251$ on WS/ER, 4.4262 on BA). In Table 1, the closed-form theory column reports this finite-time evaluation (4.425). The empirical simulation across 3,000 trials yields 4.569 ± 0.064 ($+0.144$, $+2.25\sigma$); across the 18 displayed non-zero cells in Table 1, an omnibus goodness-of-fit test ($\chi^2 = 19.2, \text{df} = 18, p = 0.38$) confirms empirical agreement with analytical predictions.

2.4 Human Error Correlation and Crossover Boundaries

In empirical human groups, individual judgments are subject to correlated estimation errors, common heuristic shortcuts, and shared information sources (Lorenz et al., 2011; Soll and Larrick, 2009; Becker, Brackbill, and Centola, 2017). We parameterize global pairwise equicorrelation in initial human priors $x_i(0)$ with $\rho_h \in [0,1]$:

$$x_i(0) = \theta + \sqrt{\rho_h}\,\eta_{h,\text{shared}} + \sqrt{1-\rho_h}\,\eta_{h,i} \tag{15}$$

where $\eta_{h,\text{shared}}, \eta_{h,i} \sim \mathcal{N}(0, \sigma_{\text{human}}^2)$ with $\sigma_{\text{human}} = 15.0$ ($\sigma_{\text{human}}^2 = 225.0$). Crucially, because $\mathbb{E}[\eta_{h,\text{shared}}] = 0$, ρ_h parameterizes zero-mean correlated estimation error, reserving systematic directional bias for μ_{AI}. The network subsequently aggregates these initial priors; human agents receive no new independent estimation signals during deliberation rounds.

The expected squared error of the unassisted human crowd mean ($\alpha = 0$) at finite horizon T evaluates to:

$$\begin{aligned}\text{MSE}_{\text{human}}(\rho_h, T) &= \sigma_{\text{human}}^2\left[\rho_h + (1-\rho_h)\|\mathbf{q}_T\|_2^2\right] \\ &= A_T + B_T\rho_h\end{aligned} \tag{16}$$

where $A_T \equiv \sigma_{\text{human}}^2\|\mathbf{q}_T\|_2^2 = 225\|\mathbf{q}_T\|_2^2$ and $B_T \equiv \sigma_{\text{human}}^2(1-\|\mathbf{q}_T\|_2^2) = 225(1-\|\mathbf{q}_T\|_2^2)$. On Watts-Strogatz small-world graphs at $T = 30$, $\|\mathbf{q}_{30}\|_2^2 \approx 0.001015$, yielding $A_{30} \approx 0.228$ and $B_{30} \approx 224.772$. This evaluates to baseline unassisted errors of 0.229 ($\rho_h = 0.00$), 4.72 ($\rho_h = 0.02$), 11.47 ($\rho_h = 0.05$), and 22.71 ($\rho_h = 0.10$).

Equating the AI-mediated collective error $E(\alpha, \rho, \rho_h, T)$ in Equations 11–13 with the unassisted baseline $\text{MSE}_{\text{human}}(\rho_h, T)$ yields the exact finite-time crossover correlation threshold $\rho_h^*(\alpha)$ for any $\alpha > 0$ (the denominator vanishes at $\alpha = 0$) above which adopting algorithmic reliance α achieves strictly lower collective error than the unassisted human crowd:

$$\rho_h^*(\alpha) = \frac{E_{\text{AI}}(\alpha, \rho, T) - A_T\left(1-(1-\alpha)^{2T}\right)}{B_T\left(1-(1-\alpha)^{2T}\right)} \tag{17}$$

where $E_{\text{AI}}(\alpha, \rho, T)$ denotes $E_{\text{AI}}^{\text{dyn}}$ or $E_{\text{AI}}^{\text{stat}}$, respectively. Evaluating Equation 17 under monopoly monoculture ($\rho = 0.90, T = 30$) on Watts-Strogatz networks reveals:

- Dynamic Noise Model: At per-step reliance $\alpha = 0.10$ ($w_{\text{cum}} = 95.8\%$), $\rho_h^* \approx 1.87\%$. At moderate per-step reliance $\alpha = 0.40$, $\rho_h^* \approx 3.28\%$. At complete reliance ($\alpha = 1.00$), $\rho_h^* = \frac{18.40-0.228}{224.772} \approx 8.1\%$ (exact 8.085%).
- Static Noise Model: At per-step reliance $\alpha = 0.10$ ($w_{\text{cum}} = 95.8\%$), $\rho_h^* \approx 7.42\%$. At moderate per-step reliance $\alpha = 0.40$, rapid saturation to the ceiling ($(1-0.4)^{30} \approx 10^{-7}$) yields $\rho_h^* \approx 8.1\%$ (8.085%). At complete reliance ($\alpha = 1.00$), $\rho_h^* \approx 8.1\%$ (8.085%).

This threshold clarifies a central tension in algorithmic monoculture: the dramatic collective degradation observed under pure monoculture ($\alpha = 1.0$) arises when benchmarking against an idealized, orthogonal human crowd ($\rho_h = 0$). Differentiating Equations 11 and 13 at $\alpha = 0$ yields $\frac{\partial E}{\partial \alpha}\big|_{\alpha=0} = -2T\,\mathrm{MSE}_{\mathrm{human}}(\rho_h) < 0$ for all $\rho_h \geq 0$, establishing that the marginal addition of AI initially decreases collective error even in orthogonal crowds (where a shallow dip exists: dynamic $\alpha^* \approx 0.0017$, error 0.218 vs. 0.229; static $\alpha^* \approx 0.0004$, a -1.2% reduction). When human crowds exhibit correlated heuristics ($\rho_h > 0$), this negative initial gradient is amplified by orders of magnitude (e.g., $-2(30)(22.71) \approx -1362.6$ at $\rho_h = 0.10$), dramatically deepening and widening the U-shaped reliance profile. Combined with the boundary condition $E(1) = 18.40 \gg \min_{\alpha\in[0,1]} E(\alpha)$ (e.g., 18.40 vs. optimal 3.82 at $\rho_h = 0.10$), this negative gradient guarantees that the social optimum α^* lies strictly in the interior $(0, 1)$. Under dynamic noise at per-step $\alpha = 0.10$, AI improves collective accuracy whenever $\rho_h > 1.87\%$, whereas under static task persistence, the AI repeats deterministic errors and improves collective accuracy at per-step $\alpha = 0.10$ only when human correlation exceeds 7.42%. Nevertheless, if reliance is calibrated to the static social optimum (per-step $\alpha^* \approx 0.008$, cumulative reliance $w_{\mathrm{cum}} \approx 20\%$), AI anchors the crowd even at modest correlated human error ($\rho_h = 0.02$), reducing collective error by 20%.

2.5 Unassisted Consensus Across Social Topologies

At $\alpha = 0$, beliefs evolve via pure DeGroot consensus $\mathbf{x}(t) = \mathbf{W}^t\mathbf{x}(0)$, converging asymptotically to $m(0) = \boldsymbol{\pi}^\top\mathbf{x}(0)$ where $\pi_i = \frac{d_i}{\sum_k d_k}$ for the lazy random-walk trust matrix $W_{ii} = 0.5, W_{ij} = 0.5/d_i$. At finite horizon T, the exact unweighted consensus variance is $\sigma^2_{\mathrm{human}}\|\mathbf{q}_T\|_2^2 = \sigma^2_{\mathrm{human}}\|\frac{1}{N}\mathbf{1}^\top\mathbf{W}^T\|_2^2$, which expands from the minimal uniform variance $\sigma^2_{\mathrm{human}}/N = 0.225$ at $T = 0$ toward the asymptotic stationary bound $\sigma^2_{\mathrm{human}}\sum_{i=1}^N \pi_i^2$ (0.229 for WS/ER, 0.557 for representative BA, ensemble expectation 0.536 ± 0.029) as T exceeds the spectral relaxation time $\tau_{\mathrm{rel}} = \frac{1}{1-\lambda_2(\mathbf{W})}$.

1. Watts-Strogatz Small-World Graphs: Regular ring lattice with random rewiring ($k = 6, p = 0.1$). Realized degrees yield $\sum \pi_i^2 \approx 0.001016$, predicting an asymptotic baseline error of $225 \times 0.001016 \approx 0.229$ (empirical at $T = 30$: 0.233 ± 0.010). Because high spatial clustering ($C \approx 0.44$; analytic $C(p) = C(0)(1-p)^3 \approx 0.437$) and moderately longer characteristic path lengths (6.1 vs. 2.0 on ER) yield a second eigenvalue of $\lambda_2(\mathbf{W}) \approx 0.984$ (spectral relaxation time $\tau_{\mathrm{rel}} = \frac{1}{1-\lambda_2} \approx 62$ rounds), spectral mixing remains incomplete at finite horizon $T = 30$, explaining why Watts-Strogatz graphs retain non-zero local diversity (1.454 ± 0.011).
2. Erdős-Rényi Random Graphs: Uniform connection probability ($p = 0.05$, $\bar{d} \approx 50$). Realized degrees yield $\sum \pi_i^2 = \frac{1}{N}(1 + \mathrm{Var}(d)/\bar{d}^2) \approx 0.001019$, predicting $225 \times 0.001019 \approx 0.229$ (empirical: 0.231 ± 0.010). Rapid spectral expansion drives diversity to 0.000 well before $T = 30$.
3. Barabási-Albert Scale-Free Graphs: Preferential attachment ($m = 3$). Hubs concentrate stationary weight: for the representative realized graph held fixed in the simulation sweep (seed 42), $\sum \pi_i^2 = 0.002475 \approx 0.00248$ and $\|\mathbf{q}_{30}\|_2^2 \approx 0.002474$ predict baseline error $225 \times 0.002474 \approx 0.557$ (seed 44 yields $0.002499 \implies 0.562$). Across 100 independently generated BA graphs, ensemble weight is $\sum \pi_i^2 = 0.00238 \pm 0.00013$ (std dev), predicting ensemble baseline 0.536 ± 0.029. In simulation, holding network structure fixed while resampling initial errors across $1,000$ trials yields 0.517 ± 0.023, matching representative theory within 1.74 SEM (1.97 SEM against 0.562; within 0.8 SEM of ensemble mean) (Golub and Jackson, 2010).

The pooled theoretical baseline across representative graphs is $(0.229 + 0.229 + 0.557)/3 \approx 0.338$ (and $(0.229 + 0.229 + 0.536)/3 \approx 0.331$ for graph ensembles), tracking the simulated pooled baseline of 0.327 ± 0.009 within 1.2 SEM.

3 Model and Simulation Methodology

3.1 Simulation Framework

We simulate networked deliberation across $N = 1,000$ agents structured on graphs $G = (V, E)$ generated via NetworkX (v3.6/3.7) in Python 3.10+ (Hagberg, Schult, and Swart, 2008). Parameter sweeps are evaluated over $T = 30$ timesteps across 193,000 unique Monte Carlo runs (199,000 total executed trials across all branches; because the seed formula incorporates the correlation arm, parallel baseline executions represent statistically independent draws) to validate analytical derivations:[1]

- Experiment 1 (Core Reliance Sweep): 10 non-zero reliance levels ($\alpha \in \{0.1, 0.2, \ldots, 1.0\}$) $\times$ 3 correlation regimes ($\rho \in \{0.00, 0.50, 0.90\}$) $\times$ 3 topologies (Watts-Strogatz, Erdős-Rényi, Barabási-Albert) $\times$ 1,000 independent trials = 90,000 runs, plus 3 topologies $\times$ 1,000 baseline trials ($\alpha = 0$) = 93,000 unique runs ($3,000$ pooled runs per (α, ρ) cell for $\alpha > 0$; 99,000 total executed trials).
- Experiment 2 (Human Correlation Sweep): Evaluated on Watts-Strogatz small-world networks across 20 fine reliance levels

[1] Replication code, simulation manifests, and analysis scripts are available at https://github.com/Ashuraj-Herode22/algorithmic-monoculture-sim.

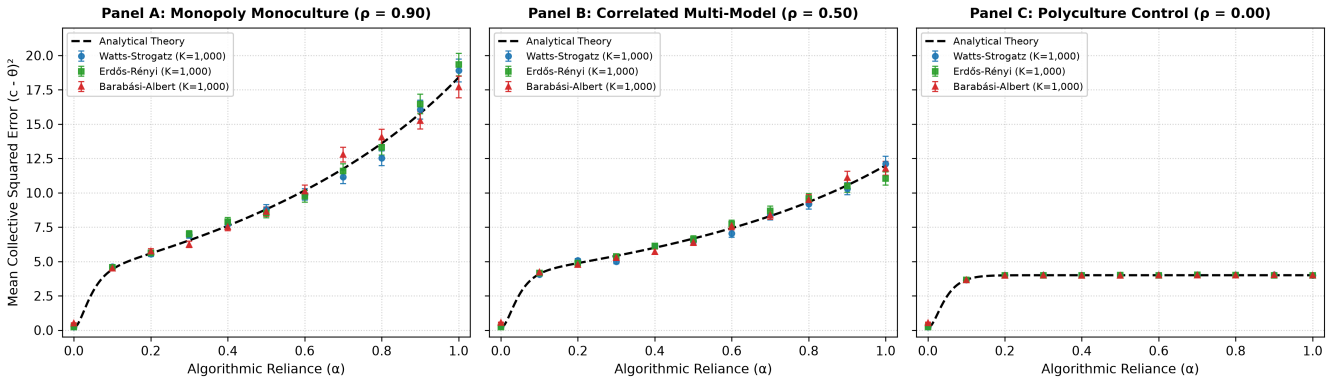


Figure 1: Collective Error vs. Algorithmic Reliance ($N = 1,000, T = 30, K = 1,000$ Trials per Condition, 99,000 Runs). Collective Squared Error $(c(T) - \theta)^2$ as a function of reliance α across independent trials for Watts-Strogatz (blue circles), Erdős-Rényi (green squares), and Barabási-Albert (red triangles) networks at $\rho_h = 0$. Dashed curves overlay the closed-form analytical prediction (Equation 11). Left (Panel A): Polyculture ($\rho = 0.00$), stabilizing at the directional bias floor ($\mu_{AI}^2 = 4.00$). Center (Panel B): Correlated Multi-Model ($\rho = 0.50$), escalating to 11.644 ± 0.293. Right (Panel C): Monopoly Monoculture ($\rho = 0.90$), escalating to 18.648 ± 0.468.

($\alpha \in [0.001, 0.10]$) $\times$ 4 human correlation levels ($\rho_h \in \{0.00, 0.02, 0.05, 0.10\}$) under monopoly monoculture ($\rho = 0.90$) $\times$ 1,000 independent trials = 80,000 runs.

- Experiment 3 (Static vs. Dynamic Persistence Sweep): Evaluated on Watts-Strogatz small-world networks across 20 fine reliance levels under static machine noise ($\rho = 0.90, \rho_h = 0.00$) $\times$ 1,000 independent trials = 20,000 runs (directly compared against the 20,000 dynamic-noise $\rho_h = 0.00$ runs from Experiment 2, evaluating 40,000 trajectory arms).

Network structures are held fixed using representative graph seeds (WS 42, ER 42, BA 42/44) while agent priors and advice vectors are independently resampled across trials; SEMs capture Monte Carlo resampling uncertainty conditional on these fixed graph realizations. Optimal reliance levels α^* are computed via Brent's scalar minimization (scipy.optimize.minimize_scalar, tolerance 10^{-6}).

Agent Prior Initialization

At $t = 0$, human agents are initialized with prior beliefs centered around true state $\theta = 100.0$:

$$x_i(0) = \theta + \sqrt{\rho_h}\,\eta_{h,\text{shared}} + \sqrt{1 - \rho_h}\,\eta_{h,i} \tag{18}$$

with $\eta_{h,\text{shared}}, \eta_{h,i} \sim \mathcal{N}(0, \sigma^2_{\text{human}})$ and $\sigma_{\text{human}} = 15.0$ ($\sigma^2_{\text{human}} = 225.0$).

Algorithmic Advice Parameterization

We emphasize that chosen parameter values ($\mu_{AI} = 2.0, \sigma_{AI} = 4.0, \sigma_{\text{human}} = 15.0, \rho = 0.90, T = 30, N = 1,000$) serve as stylized order-of-magnitude benchmarks to illustrate qualitative mechanics rather than empirical constants; specific numerical thresholds and optima are parametric functions of this setup, and per-step reliance α^* is defined relative to the $T = 30$ horizon. In our experiments, we evaluate three correlation regimes in Equation 3 ($\mu_{AI} = 2.0$, $\sigma_{AI} = 4.0$, total single-query MSE $= \mu_{AI}^2 + \sigma_{AI}^2 = 20.0$):

- $\rho = 0.90$: Monopoly Monoculture (single centralized model queried across the network; 10% residual variance captures prompt/temperature noise).
- $\rho = 0.50$: Correlated Multi-Model (competing frontier models sharing pre-training corpora and alignment pipelines).
- $\rho = 0.00$: Polyculture Control (independent, variance-decorrelated models sharing only baseline mean offset μ_{AI}).

DeGroot Trust Matrix

The row-stochastic trust matrix $\mathbf{W} \in \mathbb{R}^{N \times N}$ is constructed from graph adjacency matrix $\mathbf{A}$ by assigning self-weight $W_{ii} = 0.5$ and uniform neighbor weights $W_{ij} = 0.5/d_i$ for $j \in \mathcal{N}(i)$.

4 Experimental Results

4.1 Collective Error Escalation and Network Invariance

Figure 1 and Table 1 present the empirical simulation results across 99,000 Monte Carlo runs alongside closed-form analytical predictions.

At $\alpha = 0.0$ with orthogonal priors ($\rho_h = 0$), unassisted human deliberation achieves high collective accuracy (0.327 ± 0.009) through idiosyncratic error cancellation.

Table 1: High-Sample Simulation Results ($N = 1,000$ Agents, $T = 30$ Timesteps, $K = 1,000$ Independent Trials per Condition, 99,000 Total Runs at $\rho_h = 0$). Values denote Mean ± Standard Error of the Mean (SEM) across pooled runs (3,000 trials per reliance level across WS, ER, and BA topologies) alongside finite-time closed-form theory (E_c^{dyn}, Equation 11). Baseline $\alpha = 0.0$ rows present $K = 1,000$ trials for each individual topology and 3,000 pooled cross-topology baseline runs (BA theory reports 0.557 for representative realized graph; ensemble expectation is 0.536; pooled theory reports 0.338 representative, 0.331 ensemble).

Reliance	Polyculture ($\rho = 0.00$)		Multi-Model ($\rho = 0.50$)		Monoculture ($\rho = 0.90$)		Analytical Theory ($T = 30$)		
α	Coll. Error	Diversity	Coll. Error	Diversity	Coll. Error	Diversity	Theory ($\rho = 0$)	Theory ($\rho = 0.5$)	Theory ($\rho = 0.9$)
0.0 (WS Small-World)	Collective Error: 0.233 ± 0.010			Diversity: 1.454 ± 0.011			0.229	0.229	0.229
0.0 (ER Random)	Collective Error: 0.231 ± 0.010			Diversity: 0.000 ± 0.000			0.229	0.229	0.229
0.0 (BA Scale-Free)	Collective Error: 0.517 ± 0.023			Diversity: 0.000 ± 0.000			0.557	0.557	0.557
0.0 (Pooled Baseline)	Collective Error: 0.327 ± 0.009			Diversity: 0.485 ± 0.013			0.338	0.338	0.338
0.1	3.666 ± 0.003	0.224 ± 0.000	4.147 ± 0.048	0.112 ± 0.000	4.569 ± 0.064	0.023 ± 0.000	3.669	4.09	4.425
0.2	3.992 ± 0.003	0.808 ± 0.001	4.908 ± 0.071	0.403 ± 0.000	5.636 ± 0.102	0.081 ± 0.000	3.992	4.88	5.59
0.4	3.997 ± 0.005	2.872 ± 0.002	5.990 ± 0.113	1.436 ± 0.001	7.714 ± 0.171	0.287 ± 0.000	4.004	6.00	7.60
0.6	4.005 ± 0.006	6.037 ± 0.005	7.450 ± 0.163	3.020 ± 0.002	9.915 ± 0.248	0.603 ± 0.000	4.007	7.43	10.17
0.8	4.016 ± 0.008	10.347 ± 0.009	9.420 ± 0.218	5.174 ± 0.004	13.298 ± 0.323	1.035 ± 0.001	4.011	9.34	13.60
1.0 (Pure AI)	$\mathbf{3.997 \pm 0.009}$	$\mathbf{15.980 \pm 0.013}$	$\mathbf{11.644 \pm 0.293}$	$\mathbf{7.995 \pm 0.007}$	$\mathbf{18.648 \pm 0.468}$	$\mathbf{1.599 \pm 0.001}$	**4.016**	**12.01**	**18.40**

Disaggregating baseline performance by topology confirms our spectral derivation: Barabási-Albert scale-free networks exhibit higher consensus error (0.517 ± 0.023, predicted 0.536) than Watts-Strogatz (0.233 ± 0.010, predicted 0.229) and Erdős-Rényi (0.231 ± 0.010, predicted 0.229).

Under Monopoly Monoculture ($\rho = 0.90$), collective error escalates from baseline to 18.648 ± 0.468 at $\alpha = 1.0$. This represents a 36- to 81-fold increase depending on network topology, with an average 57-fold escalation relative to the pooled baseline (though crowd error of 18.6 remains 12× lower than unassisted individual error of 225.0 due to baseline scaling as $\mathcal{O}(1/N)$).

Crucially, this error surge is not caused by post-deliberation "diversity loss" per se: unassisted human consensus on connected Erdős-Rényi and Barabási-Albert graphs also converges to zero belief variance (Diversity$(T) = 0.000$) as agents agree, yet achieves minimal error because initial human errors cancel out during aggregation. Rather, collective error escalates under monoculture because external AI advice continuously injects non-canceling directional bias (μ_{AI}) and shared systemic error ($\sqrt{\rho}\eta_{\text{shared}}$) that shift the crowd mean trajectory $c(t)$ itself (and stationary surrogate $m(t) \equiv \boldsymbol{\pi}^\top \mathbf{x}(t)$).

Across all reliance levels $\alpha \geq 0.1$, the three network topologies collapse onto the identical analytical error trajectory (Figure 1), empirically confirming our proof that row-stochastic networks are transparent to uniform external signals. With 3,000 pooled trials per cell, empirical averages match Equation 11 across the full parameter space: at $\alpha = 0.1$, empirical error (4.569 ± 0.064) tracks theory within 2.3 standard errors (4.42); at $\alpha = 0.6$, empirical error (9.915 ± 0.248) matches theory (10.17); and at $\alpha = 1.0$, empirical error (18.648 ± 0.468) aligns with the theoretical ceiling (18.40).

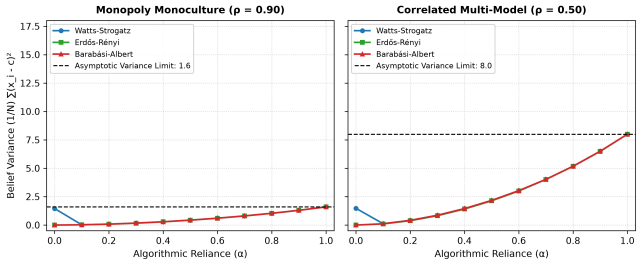


Figure 2: Belief Variance and Residual Dispersion under Algorithmic Mediation ($N = 1,000, T = 30$). Empirical crowd diversity $\frac{1}{N}\sum_i (x_i(T) - c(T))^2$ as a function of reliance α across topologies (Equation ??). Residual belief diversity scales directly with idiosyncratic variance $(1-\rho)\sigma_{AI}^2$ as $\alpha \to 1$, bounding terminal diversity at 1.60 under monopoly monoculture ($\rho = 0.90$) and 8.00 under multi-model competition ($\rho = 0.50$).

4.2 Correlated Multi-Model Competition

A common intuition in technological policy is that market competition among multiple foundation model providers will naturally prevent algorithmic monoculture. Our results (Figure 1B, Table 1) demonstrate that when competing models share overlapping web pre-training corpora and alignment protocols such that between-model error correlation remains $\rho = 0.50$, collective error still escalates to 11.644 ± 0.293 at $\alpha = 1.0$ (analytical prediction: 12.01, via finite-N ceiling $\mu_{AI}^2 + \sigma_{AI}^2[\rho + \frac{1-\rho}{N}] = 4 + 16[0.50 + 0.50/1000] = 12.008$). In an oligopolistic market where n_m agents consult provider $m \in \{1, \ldots, M\}$, the exact finite-population probability that two distinct agents consult the same provider is $p_{\text{same}} = \frac{\sum_m n_m(n_m-1)}{N(N-1)}$, yielding effective pairwise correlation $\rho_{\text{eff}} = \rho_{\text{between}} + (\rho_{\text{within}} - \rho_{\text{between}})p_{\text{same}}$ and finite-N collective MSE $E_{\text{coll}} = \mu_{AI}^2 + \sigma_{AI}^2[\frac{1}{N} + \frac{N-1}{N}\rho_{\text{eff}}]$.

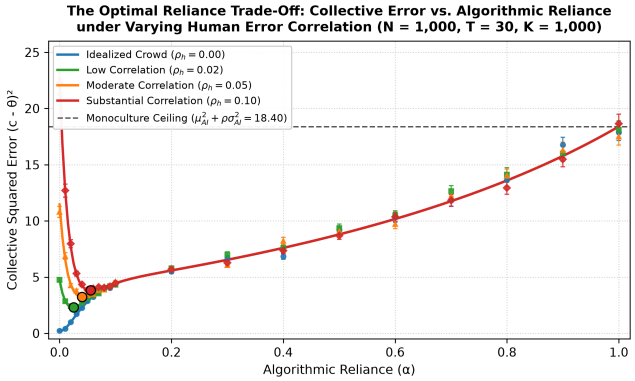


Figure 3: The Optimal Reliance Trade-Off ($N = 1,000, T = 30, K = 1,000$). Collective Squared Error vs. per-step reliance α across four human correlation regimes ($\rho_h \in \{0.00, 0.02, 0.05, 0.10\}$) on Watts-Strogatz graphs under dynamic noise ($\rho = 0.90$). Curves denote analytical theory (Equation 11); markers denote Monte Carlo trials (± SEM). Under correlated human error ($\rho_h > 0$), collective error follows a U-shaped profile: calibrated reliance ($\alpha^* \in [0.024, 0.056]$, $w_{\text{cum}} \in [51\%, 82\%]$) substantially lowers error by anchoring human bias, whereas excessive reliance ($\alpha \to 1.0$) escalates error toward the monoculture ceiling (18.40).

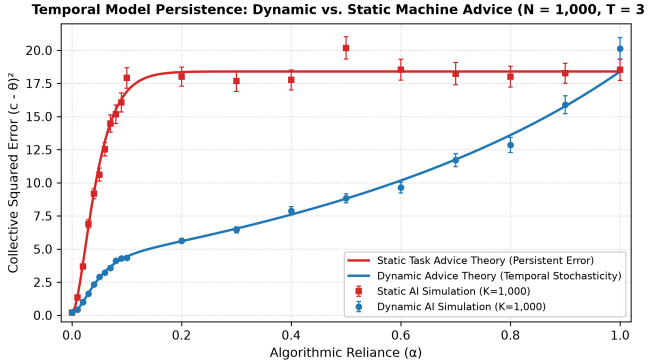


Figure 4: Temporal Model Persistence: Dynamic vs. Static Machine Advice ($N = 1,000, T = 30, K = 1,000$). Collective error under dynamic temporal stochasticity (blue circles, Equation 11) vs. persistent task-level error (red squares, Equation 13) on Watts-Strogatz graphs ($\rho = 0.90, \rho_h = 0$). Repeating deterministic mistakes eliminates multi-round variance dampening, rapidly saturating at the monoculture ceiling (18.40).

For balanced partitions ($n_m = N/M$), $p_{\text{same}} \to 1/M$ in large populations, yielding the asymptotic approximation $\rho_{\text{eff}} \approx 0.50 + \frac{0.40}{M}$ for $\rho_{\text{within}} = 0.90$ and $\rho_{\text{between}} = 0.50$. Evaluating this approximation at $\alpha = 1.0$ gives 15.20 for a duopoly ($M = 2$), 14.13 for $M = 3$, and 13.28 for $M = 5$ (interpolating smoothly between 18.40 at $M = 1$ and 12.01 as $M \to \infty$). Correlated model multiplicity thus provides limited epistemic protection.

4.3 The Polyculture Control: Independent Advice

Under the polyculture control ($\rho = 0.00$, Figure 1A), idiosyncratic machine variance cancels across the crowd and collective error stabilizes at the squared directional bias floor ($\mu_{AI}^2 = 4.00$), reaching 3.997 ± 0.009 at $\alpha = 1.0$. While variance decorrelation eliminates the $\rho\sigma_{AI}^2$ penalty, any shared directional alignment offset $\mu_{AI} \neq 0$ persists unattenuated in the crowd consensus.

4.4 Dynamic vs. Static AI Noise: Quantitative Regimes

As illustrated in Figure 4, the temporal persistence of machine errors governs the quantitative shape of the reliance trade-off. Table 2 compares the optimal social reliance α^*, percentage error reduction relative to unassisted deliberation, and stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) across realistic human correlation levels ($\rho_h \in \{0.02, 0.05, 0.10\}$).

Under dynamic noise, temporal averaging dampens shared sampling variance by $\frac{\alpha}{2-\alpha}$, yielding optimal per-step reliance $\alpha^* \in [0.024, 0.056]$ ($w_{\text{cum}} \in [51\%, 82\%]$) and reducing collective error by 51%–83%. Under static task persistence, where machine errors repeat deterministically, temporal dampening vanishes; the social optimum shifts downward to per-step $\alpha^* \in [0.008, 0.026]$ ($w_{\text{cum}} \in [20\%, 55\%]$), yielding 20%–55% error reductions.

Table 2: Optimal Social Reliance (α^*), Error Reduction vs. Unassisted Deliberation, and Stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) Under Dynamic vs. Static AI Noise ($\rho = 0.90, T = 30$). Cumulative reliance $w_{\text{cum}} \in [51\%, 82\%]$ (dynamic) and $[20\%, 55\%]$ (static). $\mathcal{R}_{\text{overshoot}}$ evaluates collective error under isolated benchmarks ($\alpha^{\text{dyn}}_{\text{isolated}} = 0.106$ dynamic; $\alpha_{\text{step}} = 0.080$ static) relative to α^*; parenthetical values report complete reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40).

Human Corr. (ρ_h)	Unassisted MSE	Dynamic Noise (Opt α^* / Cut / Ratio)	Static Noise (Opt α^* / Cut / Ratio)
$\rho_h = 0.02$	4.72	0.024 / 51% / 2.0 (8.0)	0.008 / 20% / 4.1 (4.9)
$\rho_h = 0.05$	11.47	0.041 / 72% / 1.4 (5.7)	0.016 / 38% / 2.2 (2.6)
$\rho_h = 0.10$	22.71	0.056 / 83% / 1.2 (4.8)	0.026 / 55% / 1.5 (1.8)

4.5 Private Benchmarks vs. Epistemic Overshoot

To understand why decentralized agents over-rely on foundation models, consider an isolated individual choosing reliance without social interaction:

- Static Task-Persistent Environment: For deterministic AI advice on a fixed factual query, an isolated agent minimizes:

$$\min_{\alpha} \mathbb{E}[(x_i - \theta)^2] = 225(1-\alpha)^2 + 20\alpha^2 \qquad (19)$$

yielding private optimum $\alpha^*_{\text{isolated}} = \frac{225}{245} \approx 0.918$

and dropping individual error from 225.0 to 18.37 (12.25× reduction). Over $T = 30$ rounds, retaining weight $(1 - 0.918) = 0.082$ corresponds to per-step reliance $\alpha_{\text{step}} = 1 - (1 - 0.918)^{1/30} \approx 0.080$.

- Dynamic Temporal Stochasticity: When an isolated agent queries the model with independent stochastic draws across rounds, temporal averaging minimizes:

$$\min_{\alpha} \Big[(1-\alpha)^{2T}\sigma_h^2 + \mu_{AI}^2(1-(1-\alpha)^T)^2 + \sigma_{AI}^2 \frac{\alpha(1-(1-\alpha)^{2T})}{2-\alpha}\Big] \quad (20)$$

For $T = 30$, numerical optimization yields $\alpha_{\text{isolated}}^{\text{dyn}} \approx 0.106$, driving isolated error to 4.89 (46.0× reduction over unassisted heuristics of 225.0).

Macro-Level Overshoot Under Hypothesized Correlation Neglect: When decentralized agents enter a connected social network under hypothesized correlation neglect (DeMarzo, Vayanos, and Zwiebel, 2003; Enke and Zimmermann, 2019)—treating neighbors as independent signals—they adopt private reliance benchmarks ($\alpha_{\text{step}} \approx 0.080$ static, $\alpha_{\text{isolated}}^{\text{dyn}} \approx 0.106$ dynamic) rather than internalizing the collective value of preserving independent human information.

Defining the stylized Epistemic Reliance Overshoot Ratio ($\mathcal{R}_{\text{overshoot}}$) as the counterfactual benchmark ratio of collective error under homogeneous adoption of the isolated benchmark to minimum collective error at the social optimum α^*:

$$\mathcal{R}_{\text{overshoot}} = \frac{E(\alpha_{\text{isolated}}^{\text{bench}}, \rho, \rho_h, T)}{\min_\alpha E(\alpha, \rho, \rho_h, T)} \quad (21)$$

Unlike the game-theoretic Price of Anarchy (Koutsoupias and Papadimitriou, 1999; Roughgarden and Tardos, 2002), which measures welfare degradation in Nash equilibrium, our ratio evaluates counterfactual welfare loss when boundedly rational agents adopt individually optimal reliance under hypothesized correlation neglect. While Kleinberg, Sinanaj, and Tardos (2026) prove a tight bound of 2.0 on the Price of Anarchy in matching markets under monoculture, our non-strategic ratio spans $[1.5, 4.1]$ under static persistence and $[1.2, 2.0]$ under dynamic noise (Table 2). At complete per-step reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40), the ratio reaches $[4.8, 8.0]$ (dynamic) and $[1.8, 4.9]$ (static) due to absent temporal averaging ($\frac{\alpha}{2-\alpha} = 1$).

Crucially, when human error correlation exceeds $\rho_h^* = 8.08\%$, unassisted crowd error exceeds the pure monoculture ceiling ($\text{MSE}_{\text{human}} > 18.40$). In this regime, complete reliance ($\alpha_{\text{step}} = 1.0$, MSE = 18.40) does not degrade accuracy below unassisted baseline (22.71 at $\rho_h = 0.10$), but represents an epistemic overshoot: complete reliance forfeits about 64 percentage points of error reduction (77% of achievable gains) relative to the dynamic social optimum at $\rho_h = 0.10$, rising to about 80 points ($\approx 100\%$) just above the 8.08% threshold. Decentralized adoption of the dynamic isolated benchmark ($\alpha_{\text{isolated}}^{\text{dyn}} = 0.106$, error 4.56, 79.9% reduction) forfeits 3.3 points relative to the social optimum (83.2%, Table 2).

5 Discussion and Sociotechnical Governance

5.1 Individual Auditing vs. Collective Integrity

Foundation model evaluations focus primarily on isolated single-user benchmarks. Our findings show that micro-level auditing cannot guarantee macro-level epistemic safety: a model reducing individual error by over 12-fold in static tasks (18.37 vs. 225.0) or 46-fold under dynamic sampling (4.89 vs. 225.0) simultaneously degrades collective accuracy by 57-fold in orthogonal crowds (though absolute error of 18.6 remains below individual error of 225.0 via $\mathcal{O}(1/N)$ scaling) or induces epistemic overshoot up to 4.1× at the private benchmark and 8.0× at complete reliance ($\alpha_{\text{step}} = 1.0$) in correlated crowds.

5.2 Algorithmic Diversity as Public Infrastructure

Preserving distributed collective intelligence requires treating statistical independence across AI systems as public epistemic infrastructure:

1. True Epistemic Pluralism: Training model ensembles on curated, non-overlapping data partitions and distinct alignment objectives to drive cross-model correlation $\rho \to 0$ and center directional offsets ($\mathbb{E}[\mu_{AI}] = 0$); however, independence cannot be assumed a priori and requires task-level empirical auditing.
2. Calibrated Reliance Protocols: Workflows (clinical diagnostics, forecasting, peer review) should record independent human priors before consulting AI, bounding reliance to prevent human signals from being extinguished. While our stylized benchmark yields $\alpha^* \in [0.01, 0.06]$, operational thresholds must be calibrated to domain-specific error variances, persistence, and correlation.
3. Epistemic Stubbornness vs. Passive Unassisted Nodes: In an irreducible network with non-zero AI adoption ($\mathbf{D}_\alpha = \text{diag}(\boldsymbol{\alpha}) \neq \mathbf{0}$), $(\mathbf{I} - \mathbf{D}_\alpha)\mathbf{W}$ is strictly substochastic with spectral radius $\rho_{\text{sp}}((\mathbf{I} - \mathbf{D}_\alpha)\mathbf{W}) < 1$ and fixed point $(\mathbf{I} - (\mathbf{I} - \mathbf{D}_\alpha)\mathbf{W})\mathbf{1} = \mathbf{D}_\alpha\mathbf{1}$. Setting $\alpha_i = 0$ for unassisted nodes merely delays expected bias absorption $\mathbb{E}[\mathbf{x}(t)] \to (\theta + \mu_{AI})\mathbf{1}$ over asymptotic e-folding timescale $\tau_{\text{abs}} = -1/\ln(\rho_{\text{sp}}((\mathbf{I} - \mathbf{D}_\alpha)\mathbf{W}))$; passive averaging still absorbs 100% of machine bias. Socially stubborn

nodes ($W_{ii} = 1$) consulting AI ($\alpha_i > 0$) likewise pull toward the machine estimate. Structural resilience requires epistemically stubborn human anchors combining zero AI reliance ($\alpha_i = 0$) with persistent belief anchoring—either via social insulation ($W_{ii} = 1, \alpha_i = 0$, altering network irreducibility) or Friedkin-Johnsen persistence (Friedkin and Johnsen, 1990; Acemoglu et al., 2013): $x_i(t+1) = (1-\gamma_i)\sum_j W_{ij}x_j(t) + \gamma_i x_i(0)$ ($\gamma_i > 0, \alpha_i = 0$). Provided initial priors differ in expectation from machine advice ($\mathbb{E}[x_i(0)] \neq \theta + \mu_{AI}$), anchors permanently pin the stationary expected profile away from monoculture.

6 Limitations and Future Work

First, model errors are parameterized via Gaussian residuals with scalar correlation ρ; future empirical work should estimate full covariance matrices from multi-LLM residuals (Kim et al., 2025). Second, while our dynamic and static models define bounding regimes (i.i.d. vs. fully persistent task errors), real queries may exhibit intermediate temporal autocorrelation. Third, human correlation ρ_h is modeled as global equicorrelation in initial priors; empirical networks may feature clustered correlation. Fourth, our parameters ($\mu_{AI} = 2, \sigma_{AI} = 4, \sigma_h = 15, T = 30, N = 1,000$) serve as illustrative benchmarks: while qualitative U-shaped profiles and overshoot gaps persist across parameter scaling, operational thresholds must be calibrated to domain-specific variances and horizons. Fifth, we assume $\boldsymbol{\eta}_{AI} \perp \boldsymbol{\eta}_h$; models pre-trained on human text may share positive correlation with human heuristics, further diminishing collective gains. Finally, extending DeGroot updating to degree-correlated reliance ($\alpha_i \propto d_i$) and Bayesian agents who dynamically infer correlation represents an important extension.

7 Conclusion

This study formalizes the Algorithmic Monoculture Paradox: while foundation models yield substantial accuracy gains for isolated decision-makers, uncoordinated reliance in connected networks accumulates shared machine bias and erodes crowd wisdom. Under correlated human errors ($\rho_h > 0$), collective error follows a U-shaped trajectory where calibrated per-step reliance ($\alpha^* \in [0.008, 0.056]$, cumulative weight $w_{\text{cum}} \in [20\%, 82\%]$ over $T = 30$ rounds) improves collective judgment, whereas individually rational reliance overshoots the social optimum.

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